



Student Name: \_\_\_\_\_

Teacher: \_\_\_\_\_

2023

Hurlstone Agricultural High  
School  
Year 12 Assessment Task 2

# Mathematics Extension 2

## Examiners

- Mr D Potaczala
- Mr G Huxley
- Mr G Rawson

## General Instructions

- Reading time – 3 minutes
- Working time – 40 minutes
- Write using black or blue pen
- NESA-approved calculators may be used
- A Reference sheet is provided for your use
- An A4 handwritten summary sheet may be used
- In Questions 7 to 9, show relevant mathematical reasoning and/or calculations

## Total marks: 30

### Section I – 6 marks (page 2–3)

- Attempt Questions 1–6
- Allow about 8 minutes for this section

### Section II – 24 marks (pages 4–6)

- Attempt Questions 7–9
- Allow about 32 minutes for this section
- For each question use a separate answer booklet

## Section I

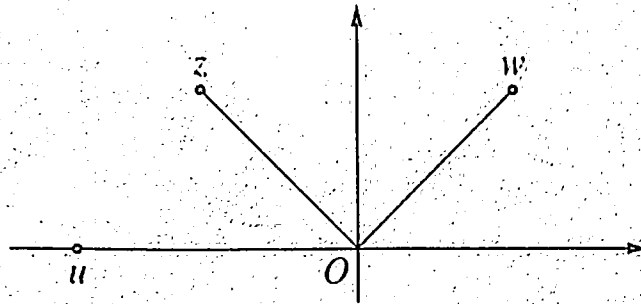
6 marks

Attempt Questions 1 to 6

Allow about 8 minutes for this section

Use the multiple-choice answer sheet for Questions 1 to 6

1. The argand diagram shows the complex numbers  $w$ ,  $z$ , and  $u$ , where  $w$  lies in the first quadrant,  $z$  lies in the second quadrant and  $u$  lies on the negative real axis.



Which statement could be true?

- A.  $u = zw$  and  $u = z + w$       B.  $u = zw$  and  $u = z - w$
- C.  $z = uw$  and  $u = z + w$       D.  $z = uw$  and  $u = z - w$
2. It is known that a particular complex number  $z$  is NOT a real number.
- Which of the following could be true for this number  $z$ ?
- A.  $\bar{z} = iz$       B.  $\bar{z} = |z^2|$
- C.  $\operatorname{Re}(iz) = \operatorname{Im}(z)$       D.  $\operatorname{Arg}(z^3) = \operatorname{Arg}(z)$
3. Which complex number lies in the region  $2 < |z - 1| < 3$ ?

A.  $1 + \sqrt{3}i$

B.  $1 + 3i$

C.  $2 + i$

D.  $3 - i$

4. Given that  $z = 1 - i$ , which expression is equal to  $z^3$ ?

A.  $\sqrt{2} \left( \cos \left( \frac{-3\pi}{4} \right) + i \sin \left( \frac{-3\pi}{4} \right) \right)$

B.  $2\sqrt{2} \left( \cos \left( \frac{-3\pi}{4} \right) + i \sin \left( \frac{-3\pi}{4} \right) \right)$

C.  $\sqrt{2} \left( \cos \left( \frac{3\pi}{4} \right) + i \sin \left( \frac{3\pi}{4} \right) \right)$

D.  $2\sqrt{2} \left( \cos \left( \frac{3\pi}{4} \right) + i \sin \left( \frac{3\pi}{4} \right) \right)$

5. Which statement is true?

A.  $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$  such that  $xy = 6$ .

B.  $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$  such that  $x + y = 6$ .

C.  $\exists x \in \mathbb{R}$  such that  $\forall y \in \mathbb{R}, x + y = 6$ .

D.  $\exists x \in \mathbb{R}$  such that  $\forall y \in \mathbb{R}, xy = 6$ .

6. Consider the statement  $P$ .

$P$ : For all integers  $n \geq 1$ , if  $n$  is a prime number then  $\frac{n(n+1)}{2}$  is a prime number.

Which of the following is true about this statement and its converse?

A. The statement  $P$  and its converse are both true.

B. The statement  $P$  and its converse are both false.

C. The statement  $P$  is true and its converse is false.

D. The statement  $P$  is false and its converse is true.

~ END OF SECTION I ~

## Section II

18 marks

Attempt Questions 7 to 9

Allow about 32 minutes for this section

Answer each question in a new answer booklet.

All necessary working should be shown in every question.

---

### Question 7 (8 marks)

Marks

- (a) Let  $z = 5 - i$ .
- (i) Find  $z + 2\bar{z}$ . 1
- (ii) Find  $\frac{i}{z}$  in the form  $x + iy$ . 1
- (b) Prove that  $2i$  is a root of  $x^3 - 3x^2 + 4x - 12 = 0$ . Hence, find all other roots. 3
- (c) Given the complex number  $z = e^{i\theta}$ , show that  $w = \frac{z^2 - 1}{z^2 + 1}$  is purely imaginary. 3

~ End of question 7 ~

Question 8 (8 marks)

Marks

- (a) (i) Sketch the region on the Argand diagram where the following inequalities hold simultaneously: 3

$$|z - 1| < \sqrt{2} \text{ and } 0 \leq \arg(z + i) \leq \frac{\pi}{4}$$

- (ii) Let  $w$  be the complex number of minimum modulus satisfying the inequalities in part (i). Express  $w$  in the form  $x + iy$ , where  $x$  and  $y$  are real number. 1

- (b) Let  $z = \cos \theta + i \sin \theta$ .

- (i) Show that  $z^n - \frac{1}{z^n} = 2i \sin n\theta$ . 2

- (ii) Using the expansion of  $\left(z - \frac{1}{z}\right)^5$ , show that: 2

$$\sin^5 \theta = \frac{\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta}{16}$$

~ End of question 8 ~

Question 9 (8 marks)

Marks

- (a) Consider the following statement, which is of the form  $P \rightarrow Q$ .

"If the pet is a stallion, then it must be a horse."

- |       |                                            |   |
|-------|--------------------------------------------|---|
| (i)   | Write the converse of the statement.       | 1 |
| (ii)  | Write the contrapositive of the statement. | 1 |
| (iii) | Write the negation of the statement        | 1 |

- (b) Write a proof by contradiction to show that: 2

There are no integers  $a$  and  $b$  such that  $21a + 3b = 1$ .

- (c) (i) Write an inequality that describes the statement: 1

The area of a rectangle cannot be greater than the square of the average length of its two sides.

- (ii) Prove the statement in part (i).

~ END OF SECTION II ~

~ END OF PAPER ~

1)

Since  $z$  and  $w$  have positive imaginary components  $u = z - w$  can be true as the imaginary parts can cancel to give only a real.

Also, can be true  $u = zw$  as the addition of an acute and obtuse argument can result in a straight angle.

**Option B**

2)

Since  $z$  is not real, we have either  $z = x + yi$  or  $z = yi$

B is not an option as the modulus is a real number

C is not an option as  $\operatorname{Re}(z) = 0$  is a possibility.

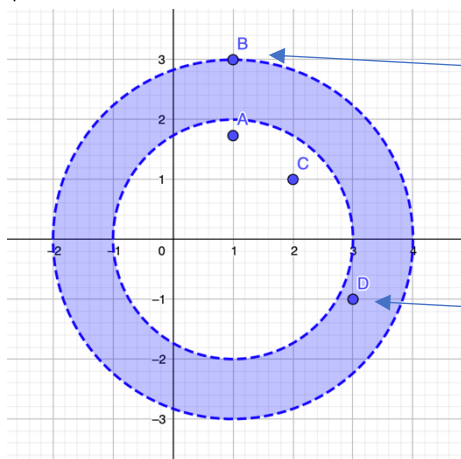
D  $\arg(r^3 e^{3i\theta}) = \arg(re^{i\theta}) \rightarrow 3i\theta = i\theta$  which is not true

Hence option A must be true ( $\bar{z} = iz$  is true if  $z = y + yi$ ).

**Option A**

3)

$$2 < |z - 1| < 3$$



It *can't* be B, because B is on the boundary, which is not included.

That's your answer, there.

D

**Option D**

4)

$$\begin{aligned}
 z^3 &= (1 - i)^3 \\
 &= \left( \sqrt{2} e^{-i\frac{\rho}{4}} \right)^3 \\
 &= 2\sqrt{2} e^{-i\frac{3\rho}{4}} \\
 &= 2\sqrt{2} \left( \cos\left(\frac{-3\rho}{4}\right) + i \sin\left(\frac{-3\rho}{4}\right) \right)
 \end{aligned}$$

### Option B

5)

Interpretations in words:

A. For all real  $x$ , there is a  $y$ -value such that  $xy=6$ . This is false, with  $x=0$  a counter-example.

B. For all real  $x$ , there is a  $y$ -value such that  $x + y = 6$ . This is true.

C. There exists a real  $x$ -value that has all  $y$ -values satisfying  $x + y=6$ . This is false because every  $x$ -value has a single unique  $y$ -value that makes the equation true.

D. There exists a real  $x$ -value that has all  $y$ -values satisfying  $xy=6$ . This is false, because every  $x$ -value in the domain has a single unique  $y$ -value that makes the equation true.

### Option B

6)

The 4 response choices consider the combinations of the statement and its converse. Consider each statement:

Statement P is false. When testing prime numbers, test the number 2 (the only even prime) and an odd prime.  $n=3$  is a counter-example to the statement.

When testing statement P you may have noticed that  $n(n+1)$  is always an even product of two factors. The only time the numerator is fully factorised by the denominator is when  $n=2$ .

Hence, the only time the fraction is prime is when  $n=2$ .

The converse: "If the fraction is prime, then  $n$  is prime" is true.

### Option D

| 2023 Yr12 HSC Assessment Task 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Question 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Solutions and Marking Guidelines                                                                                                                                                                                           |
| <b>Outcomes Addressed in this Question</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                            |
| MEX12-4<br>uses the relationship between algebraic and geometric representations of complex numbers and complex number techniques to prove results, model and solve problems                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                            |
| Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marking Guidelines                                                                                                                                                                                                         |
| a i) $z + 2\bar{z} = 5 - i + 2(5 + i)$ $= 15 + i$<br>ii) $\frac{i}{z} = \frac{i}{5 - i} = \frac{(5 + i)i}{25 + 1}$ $= \frac{-1 + 5i}{26} = -\frac{1}{26} + \frac{5}{26}i$<br>b) $f(2i) = (2i)^3 - 3(2i)^2 + 4(2i) - 12$ $= -8i + 12 + 8i - 12$ $= 0$<br>By factor theorem $2i$ is a root.<br>Since the coefficients of the polynomial are real, the complex roots come in conjugate pairs. Hence $-2i$ is another root.<br>Summing the roots we have: $2i - 2i + \alpha = -\frac{-3}{1}$ $\alpha = 3$ | 1 mark<br>Correct solution<br><br>1 mark<br>Correct solution<br><br>3 marks<br>Correct solution with clear reasoning<br><br>2 marks Substantial progress towards solution<br><br>1 mark<br>Some progress towards solution. |

c)

$$w = \frac{z^2 - 1}{z^2 + 1}$$

$$= \frac{z \left( z - \frac{1}{z} \right)}{z \left( z + \frac{1}{z} \right)}$$

$$= \frac{e^{i\theta} - e^{-i\theta}}{e^{i\theta} + e^{-i\theta}}$$

$$= \frac{\cos\theta + i\sin\theta - \cos\theta + i\sin\theta}{\cos\theta + i\sin\theta + \cos\theta - i\sin\theta}$$

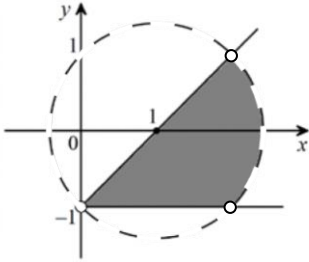
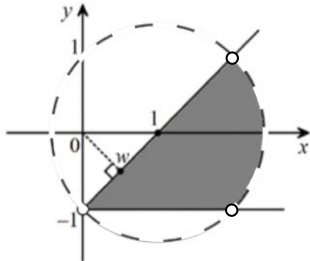
$$= \frac{2i\sin\theta}{2\cos\theta}$$

Which is imaginary.

3 marks  
Correct solution with  
clear reasoning

2 marks Substantial  
progress towards  
solution

1 mark  
Some progress  
towards solution.

| Year 12                                    | Mathematics Extension 2                                                                                                                                                                                                                                                                                                                                                      | Ass Task 2 2023 HSC                                                                                                                                                                                             |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Question No. 8                             | Solutions and Marking Guidelines                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                 |
| <b>Outcomes Addressed in this Question</b> |                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                 |
| <b>MEX12-4</b>                             | uses the relationship between algebraic and geometric representations of complex numbers and complex number techniques to prove results, model and solve problems                                                                                                                                                                                                            |                                                                                                                                                                                                                 |
| Part / Outcome                             | Solutions                                                                                                                                                                                                                                                                                                                                                                    | Marking Guidelines                                                                                                                                                                                              |
| (a)(i)                                     |                                                                                                                                                                                                                                                                                             | <u>3 marks</u> : correct solution<br><br>Note: the open circle at (0, -1) is necessary for full marks.<br><br><u>2 marks</u> : substantially correct solution<br><br><u>1 mark</u> : partially correct solution |
| (a)(ii)                                    | <p>For <math>w</math> to lie in the region in part (i) and be of minimum modulus, <math>w</math> must be in the position as shown below... ie the perpendicular to <math>y = x - 1</math>, through <math>O</math>.</p>  <p>By inspection, <math>w = \frac{1}{2} - \frac{1}{2}i</math></p> | <u>1 mark</u> : correct solution                                                                                                                                                                                |
| (b)(i)                                     | $z^n - \frac{1}{z^n} = z^n - z^{-n}$ $= (\cos q + i \sin q)^n - (\cos q + i \sin q)^{-n}$ $= \cos nq + i \sin nq - (\cos(-nq) + i \sin(-nq)) \quad (\text{de Moivre})$ $= \cos nq + i \sin nq - (\cos nq - i \sin nq)$ $= 2i \sin nq$                                                                                                                                        | <u>2 marks</u> : correct solution<br><br><u>1 mark</u> : substantially correct solution                                                                                                                         |

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| (b)(ii) | $\left(z - \frac{1}{z}\right)^5 = z^5 - 5z^4 \frac{1}{z} + 10z^3 \frac{1}{z^2} - 10z^2 \frac{1}{z^3} + 5z \frac{1}{z^4} - \frac{1}{z^5}$ $= z^5 - 5z^3 + 10z - 10\left(\frac{1}{z}\right) + 5\left(\frac{1}{z^3}\right) - \frac{1}{z^5}$ $= \left(z^5 - \frac{1}{z^5}\right) - 5\left(z^3 - \frac{1}{z^3}\right) + 10\left(z - \frac{1}{z}\right)$ <p>so, <math>(2i \sin q)^5 = 2i \sin 5q - 5(2i \sin 3q) + 10(2i \sin q)</math></p> $32i \sin^5 q = 2i \sin 5q - 10i \sin 3q + 20i \sin q$ $= \frac{2i \sin 5q - 10i \sin 3q + 20i \sin q}{32i}$ $= \frac{\sin 5q - 5 \sin 3q + 10 \sin q}{16}$ | <p><u>2 marks</u> : correct solution</p> <p><u>1 mark</u> : substantially correct solution</p> |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|

| Year 12                                                                                                      | Mathematics Extension 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Ass Task 2 2023 HSC                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Solutions and Marking Guidelines <b>Question 9</b>                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                 |
| <b>Outcomes Addressed in this Question</b>                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                 |
| E2: Chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                 |
| Part                                                                                                         | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marking Guidelines                                                                                                                              |
| <b>Q9 (a)</b>                                                                                                | <p>(a) The question has a <math>P \Rightarrow Q</math> statement.</p> <p>(i) Converse has the form: <math>Q \Rightarrow P</math></p> <p>Hence: “If the pet is a horse, it must be a stallion.”</p> <p>(ii) Contrapositive has the form: <math>\neg Q \Rightarrow \neg P</math>.</p> <p>Hence: “If the pet is not a horse, it is not a stallion.”</p> <p>(iii) Negation has the form <math>\neg P \Rightarrow \neg Q</math>.</p> <p>Hence: “If not a stallion, then the pet is not a horse.”</p> <p>Negations can also replace universal statements with “there exists”.</p> <p>Hence the alternative: “There exists a stallion that is not a horse.”</p> | <p><b>(a)(i)1 mark:</b> correct statement.</p> <p><b>(ii)1 mark:</b> correct statement.</p> <p><b>(iii)1 mark:</b> correct statement.</p>       |
|                                                                                                              | <p><b>(b)</b></p> <p>(b) Assume that <math>a</math> and <math>b</math> are integers, and</p> $21a + 3b = 1$ <p>Then: <math>3(7a + 3b) = 1</math></p> $7a + b = \frac{1}{3}$ <p>But LHS is an integer for integers <math>a</math> and <math>b</math>.</p> <p>RHS is not an integer, and so we have a contradiction.</p> <p>Hence, the assumption is untrue; both <math>a</math> and <math>b</math> can't be integers.</p>                                                                                                                                                                                                                                 | <p><b>(b)2 marks:</b> Proper set up and proof by contradiction statement included.</p> <p><b>1 mark:</b> One component of proof incomplete.</p> |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                           |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| (c) | <p>(c) (i) Let the length and width of a rectangle be <math>x, y</math>.</p> <p>Inequality will be: <math>xy \leq \left(\frac{x+y}{2}\right)^2</math></p> <p>(ii) <u>Method 1: Consider the difference:</u></p> $  \begin{aligned}  xy - \left(\frac{x+y}{2}\right)^2 &= xy - \frac{x^2 + 2xy + y^2}{4} \\  &= \frac{4xy - x^2 - 2xy - y^2}{4} \\  &= \frac{-x^2 + 2xy - y^2}{4} \\  &= -\frac{(x-y)^2}{4} \\  &\leq 0 \text{ due to the square being non-negative.}  \end{aligned}  $ $\therefore xy \leq \left(\frac{x+y}{2}\right)^2$ <p><u>Method 2: Using the AM/GM inequality.</u></p> <p>AM/GM: <math>\frac{a+b}{2} \geq \sqrt{ab}</math>      square both sides</p> $\left(\frac{a+b}{2}\right)^2 \geq ab$ <p>Which is equivalent to the statement in (i).</p> <p><i>Note: Method 2 begins with a factual statement as its “axiom”, then progresses towards the statement that needs to be proved.</i></p> | <p><b>(c)(i) 1 mark:</b> Correct answer</p> <p><b>(ii) 2 marks:</b> Proper set up and proof.</p> <p><b>1 mark:</b> One component of proof incomplete.</p> |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|